

DESIGN SYSTEM AUDIT

Computis v2.0.4 — Dev Draft Review

Full codebase audit · client-side static analysis

Date	March 29, 2023	Stage	v2.0.4 · Draft 1
Components	111+ reviewed	Issues	34 found
Scope	49 components · 14 dirs	Audit Score	68 / 100

Overall Impression

The foundation is strong — a well-architected token system (CSS variables → Tailwind → CVA), consistent file and export naming, and solid accessibility primitives from Radix UI.

However, a significant gap exists between the design system's **intent** and its **implementation**: roughly 40% of feature components bypass the token system entirely, using hardcoded Tailwind color classes and arbitrary pixel values. For a final polish pass, the biggest opportunity is closing this gap to make the system truly consistent and themeable.

Usability

Finding	Severity	Recommendation
Duplicate Button and Input components create consumer confusion (button.tsx vs enhanced-button.tsx, input.tsx vs enhanced-input.tsx)	Critical	Merge into single components. enhanced-* versions bypass the token system with hardcoded pixel sizing.
Two toast systems running simultaneously (Radix Toaster + Sonner) in App.tsx	Moderate	Consolidate to one. Both creates inconsistent notification UX and inflates bundle size.
Confidence filter buttons in TransactionsContent use hardcoded color classes instead of semantic Badge variants	Moderate	Replace with Badge variant=success / warning / error — these already exist in the system.
Icon-only buttons throughout dashboard (MoreHorizontal, menu triggers) lack aria-label attributes	Moderate	Add aria-label=More options or equivalent to every icon-only Button.

Visual Hierarchy

What draws the eye first: Page titles and large metric displays (text-display-lg) — correct for a data-dense financial dashboard.

Reading flow: Title → KPI cards → Data tables → Sidebar navigation. Well-structured, though some components break it by using raw Tailwind sizes (text-2xl, text-lg) instead of the semantic scale (text-heading-lg, text-display-sm).

Emphasis concerns

Area	Issue
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Dashboard cards	Trend indicators use text-green-600 / text-red-600 (hardcoded) rather than text-success / text-error tokens. Functional but not themeable.
Sidebar active state	Active items use bg-sidebar-accent text-white — good. But inactive text is hardcoded text-[#a3a3a3] (9 instances) instead of a token.
Chart sections	10+ raw hex values for segment colors in enhanced-pie-charts.tsx and pie-chart-sections.tsx. Won't adapt to dark mode or theme changes.

Consistency

Token Adherence — Color

Element	Issue	Recommendation
Hardcoded Tailwind colors (bg-red/green/blue-*)	~30 instances bypass semantic tokens	Replace with bg-success, bg-error, bg-warning, text-success-text — all already defined in global.css and Tailwind config.
Sidebar muted text #a3a3a3	Hardcoded hex repeated 9 times (sidebar.tsx)	Create --sidebar-muted token, use text-sidebar-muted utility.
Chart segment colors	10+ inline hex values	Move to --chart-* tokens (6 already exist — extend the set).
Modal overlays bg-black/80	Hardcoded in 4 components	Create --overlay token, e.g. hsl(0 0% 0% / 0.8).

Token Adherence — Spacing

Element	Issue	Recommendation
Arbitrary bracket values (w-[120px], max-w-[160px])	~25 instances across tables and dashboard components	For recurring widths (120/140/160px), add named width tokens to Tailwind config.
Drawer radius rounded-t-[10px]	Doesn't match radius token scale (drawer.tsx)	Use rounded-t-lg (--radius-lg) instead.
Mixed spacing approaches	p-standard vs p-4 vs p-[16px] — all meaning 16px	Pick one convention: p-4 for simple cases, p-standard for semantic density contexts.

Token Adherence — Z-Index

Value	Location	Recommendation
z-[1]	navigation-menu.tsx	Define --z-dropdown: 50
z-[100]	toast.tsx	Define --z-toast: 200
z-[9998], z-[9999]	Various overlays	Define --z-overlay: 300
z-50	enhanced-button.tsx, header	Align with token scale

Variant Parity Across Components

Variant	Badge	Button	Alert	Input
success	Yes	Yes	No	Enhanced only
warning	Yes	Yes	No	Enhanced only
error	Yes	No	No	Enhanced only

info	Yes	No	No	No
neutral	Yes	No	No	No

Recommendation: Either promote error, info, and neutral to Button and Alert, or document why Badge intentionally diverges.

Documentation vs Implementation Drift

Token	global.css (source of truth)	Documentation files	Match
Primary color	218 91% 45% (#2563EB)	218 91% 48% (lighter)	No
Accent color	45 90% 51% (Gold)	210 40% 96.1% (gray)	No
Button height sm	32px	36px (spacing.json)	No
Button height md	36px	44px (spacing.json)	No
Button height lg	44px	48px (spacing.json)	No
Typography scale	All 10 sizes	All 10 sizes	Yes
Gray scale	50–900	50–900	Yes

Developers referencing documentation files will build against incorrect values. Reconcile immediately.

Accessibility

Color Contrast

The semantic color system in global.css is WCAG 2.1 AA compliant — all *-text on *-bg combinations meet the 4.5:1 ratio. However, feature components using hardcoded Tailwind colors (e.g. text-green-600 on bg-green-100) have not been verified. The standalone HTML pages use their own palettes entirely outside the token system.

Touch Targets

Button sizes (sm 32px, md 36px, lg 44px) are documented correctly. The sm size (32px) falls below the 44px WCAG 2.5.5 minimum — the system correctly notes this should be desktop-only, but no enforcement mechanism exists.

Semantic HTML

The React app has good landmark structure: nav (sidebar), header (top bar with role=banner), main (content via SidebarInset). However, heading hierarchy is inconsistent — some pages use h1 correctly, while others use p className=text-display-lg for primary headings, breaking the document outline.

Missing Accessibility Attributes

Issue	Components	Fix
Icon-only buttons without aria-label	dashboard-cards.tsx, dashboard-content.tsx, pie-chart-sections.tsx	Add descriptive labels
Badge / Skeleton / Calendar missing forwardRef	badge.tsx, skeleton.tsx, calendar.tsx	Add React.forwardRef wrapper and displayName
HTML page tables missing aria-label	Both standalone HTML files	Add aria-label=Transaction data or similar
Confidence slider missing aria-describedby	sliding-confidence-transactions.html	Link tooltip content via aria-describedby

Standalone HTML Pages vs React App

The two standalone HTML pages (binary-automation-transactions.html and sliding-confidence-transactions.html) operate entirely outside the design system — different color palettes, typography scales, spacing, and responsive breakpoints. If these are production pages, they need to be migrated to the React component system or, at minimum, adopt the same CSS custom properties.

What Works Well

The audit identified several areas of genuine strength that should be preserved and built on:

- **Token architecture is sound.** CSS variables → Tailwind config → CVA variants is the right pattern. The foundation supports theming, dark mode, and density adjustments.
- **File and export naming is 100% consistent.** All 52 UI files use kebab-case; all exports use PascalCase.
- **cn() utility usage is universal.** Every component that needs class merging uses it correctly.
- **Typography token scale is perfectly aligned** across docs, config, and CSS — the one category with zero drift.
- **Accessibility primitives are solid.** Radix UI provides keyboard navigation, focus management, and ARIA out of the box. The header (role=banner, aria-label) and search (role=search) are well-labeled.
- **Focus-visible states** are properly implemented with ring-2 across interactive elements.
- **Reduced motion and high contrast** media queries are defined in global.css.
- **Responsive sidebar** transitions cleanly between expanded, collapsed, and mobile sheet states.

Priority Recommendations

● P0 — Fix Before Ship

1. **Consolidate duplicate components.** Merge button.tsx / enhanced-button.tsx and input.tsx / enhanced-input.tsx. The enhanced versions bypass the token system with hardcoded sizing and Tailwind color classes. One component each, using design tokens for all variants and sizes.
2. **Reconcile documentation with implementation.** Primary color, accent color, and all three button heights are wrong in the docs. Update colors.css, spacing.json, and any generated docs to match global.css (the source of truth).
3. **Replace hardcoded colors in feature components.** The ~30 instances of raw Tailwind colors should use semantic tokens (bg-success-bg, text-error-text). Single highest-impact change for consistency and dark mode readiness.

● P1 — Fix Soon

4. **Extract chart colors into tokens.** Create --chart-processed, --chart-success, etc. in global.css and map through Tailwind. Eliminates 10+ inline hex values.
5. **Create sidebar muted token.** Replace 9 instances of #a3a3a3 with text-sidebar-muted.
6. **Add aria-label to all icon-only buttons.** Audit every Button size=icon and ensure a descriptive label.
7. **Define z-index scale.** Create 4–5 tokens (--z-base, --z-dropdown, --z-modal, --z-toast, --z-overlay) and replace all arbitrary values.
8. **Consolidate to one toast system.** Pick either Radix Toaster or Sonner, not both.

● P2 — Improve Over Time

9. **Tokenize recurring arbitrary widths.** Add named width utilities for 120px, 140px, 160px to Tailwind config.
10. **Standardize heading hierarchy in React.** Ensure every page has exactly one h1, and that visual size classes don't replace semantic heading elements.

11. **Migrate standalone HTML pages.** Convert binary-automation-transactions.html and sliding-confidence-transactions.html to React components, or share the CSS custom properties from global.css.
12. **Document dark mode accent behavior.** The accent shifts from gold to gray in dark mode — currently undocumented and may confuse implementers.

Audit performed against Computis Design System v2.0. Findings based on static code analysis of all client-side files.